Application Note

Commands and Subcommands

ABSTRACT

The GD30BM2016 device supports direct commands and subcommands. Direct commands are single-byte, while word commands are two-byte. Direct commands are accessed using a 7-bit command address sent from the host via the device's serial communication interface, triggering operations, providing data values to be written to the device, or instructing the device to report data to the host. Subcommands are additional commands accessed indirectly through a 7-bit command address space and provide block data transfer functionality.

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1 Direct Commands

Base Address: 0x0400

1.1 Control Status Register

Address Offset: 0x00

Field	Bits	Default	Type	Field Description
auto_load_timeout	[3:3]	0x0	LH	0 = auto load not timeout; 1 = auto load timeout; write 1 clr.
deepsleep	[2:2]	0x0	RO	This bit indicates whether or not the device is in DEEPSLEEP mode. 0 = Device is not in DEEPSLEEP mode. 1 = Device is in DEEPSLEEP mode.
ld_timeout	[1:1]	0x0	RO	This bit is set when the Load Detect function has timed out and checking has stopped. 0 = Load Detect function has not timed out or is inactive. 1 = Load Detection function timed out and was deactivated.
ld_on	[0:0]	0x0	RO	This bit indicates whether or not the Load Detect pullup was active during the previous LD pin voltage measurement. 0 = LD pullup was not active during the previous LD pin measurement. 1 = LD pullup was active during the previous LD pin measurement.

1.2 Safety Alert A Register

Address Offset: 0x02

Field	Bits	Default	Type	Field Description
safety_alert_a	[7:0]	0x0	RO	bit7: Short Circuit in Discharge Protection bit6: Overcurrent in Discharge 2nd Tier Protection bit5: Overcurrent in Discharge 1st Tier Protection bit4: Overcurrent in Charge Protection bit3: Cell Overvoltage Protection bit2: Cell Undervoltage Protection 0 = Alert is not triggered. 1 = Alert is triggered.

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1.3 Safety Status A Register

Address Offset: 0x03

Field	Bits	Default	Type	Field Description
safety_status_a	[7:0]	0x0	RO	bit7: Short Circuit in Discharge Protection bit6: Overcurrent in Discharge 2nd Tier Protection bit5: Overcurrent in Discharge 1st Tier Protection bit4: Overcurrent in Charge Protection bit3: Cell Overvoltage Protection bit2: Cell Undervoltage Protection 0 = Fault is not triggered. 1 = Fault is triggered.

1.4 Safety Alert B Register

Address Offset: 0x04

Field	Bits	Default	Type	Field Description
safety_alert_b	[7:0]	0x0	RO	bit7: FET Overtemperature bit6: Internal Overtemperature bit5: Overtemperature in Discharge bit4: Overtemperature in Charge bit2: Internal Undertemperature bit1: Undertemperature in Discharge bit0:Undertemperature in Charge 0 = Alert is not triggered. 1 = Alert is triggered.

1.5 Safety Status B Register

Address Offset: 0x05

Field	Bits	Default	Type	Field Description
safety_status_b	[7:0]	0x0	RO	bit7: FET Overtemperature bit6: Internal Overtemperature bit5: Overtemperature in Discharge bit4: Overtemperature in Charge bit2: Internal Undertemperature bit1: Undertemperature in Discharge bit0: Undertemperature in Charge 0 = Fault is not triggered. 1 = Fault is triggered.

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1.6 Safety Alert C Register

Address Offset: 0x06

Field	Bits	Default	Type	Field Description
safety_alert_c	[7:0]	0x0	RO	bit7: Overcurrent in Discharge 3rd Tier Protection bit6: Short Circuit in Discharge Latch bit5: Overcurrent in Discharge Latch bit4: Cell Overvoltage Latch bit3: Precharge Timeout Suspend

1.7 Safety Status C Register

Address Offset: 0x07

Field	Bits	Default	Type	Field Description
safety_status_c	[7:0]	0x0	RO	bit7: Overcurrent in Discharge 3rd Tier Protection bit6: Short Circuit in Discharge Latch bit5: Overcurrent in Discharge Latch bit4: Cell Overvoltage Latch bit2: Precharge Timeout bit1:Host Watchdog Fault 0 = Fault is not triggered. 1 = Fault is triggered.

1.8 PF Alert A Register

Address Offset: 0x0A

Field	Bits	Default	Type	Field Description
pf_alert_a	[7: 0]	0x0	RO	bit7: Copper Deposition Permanent Fail bit6: Safety Overtemperature FET Permanent Fail bit4: Safety Overtemperature Permanent Fail bit3: Safety Overcurrent in Discharge Permanent Fail bit2: Safety Overcurrent in Charge Permanent Fail bit1: Safety Cell Overvoltage Permanent Fail bit0: Safety Cell Undervoltage Permanent Fail 0 = Alert is not triggered. 1 = Alert is triggered.

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1.9 PF Status A Register

Address Offset: 0x0B

Field	Bits	Default	Type	Field Description
pf_status_a	[7:0]	0x0	RO	bit7: Copper Deposition Permanent Fail bit6: Safety Overtemperature FET Permanent Fail bit4: Safety Overtemperature Permanent Fail bit3: Safety Overcurrent in Discharge Permanent Fail bit2: Safety Overcurrent in Charge Permanent Fail bit1: Safety Cell Overvoltage Permanent Fail bit0: Safety Cell Undervoltage Permanent Fail 0 = Fault is not triggered. 1 = Fault is triggered.

1.10 PF Alert B Register

Address Offset: 0x0C

Field	Bits	Default	Type	Field Description
pf_alert_b	[7:0]	0x0	RO	bit7: Short Circuit in Discharge Latch Permanent Fail bit4: Voltage Imbalance Active Permanent Fail bit3: Voltage Imbalance at Rest Permanent Fail bit2: Second Level Protector Permanent Fail bit1: Discharge FET Permanent Fail bit0: Charge FET Permanent Fail 0 = Alert is not triggered. 1 = Alert is triggered.

1.11 PF Status B Register

Address Offset: 0x0D

Field	Bits	Default	Type	Field Description
pf_status_b	[7:0]	0x0	RO	bit7: Short Circuit in Discharge Latch Permanent Fail bit4: Voltage Imbalance Active Permanent Fail bit3: Voltage Imbalance at Rest Permanent Fail bit2: Second Level Protector Permanent Fail bit1: Discharge FET Permanent Fail bit0: Charge FET Permanent Fail 0 = Fault is not triggered. 1 = Fault is triggered.

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1.12 PF Alert C Register

Address Offset: 0x0E

Field	Bits	Default	Type	Field Description
pf_alert_c	[7:0]	0x0	RO	bit6: Hardware Mux Permanent Fail bit5: Internal VSS Measurement Permanent Fail bit4: Internal Voltage Reference Permanent Fail bit3: Internal LFO Permanent Fail 0 = Alert is not triggered. 1 = Alert is triggered.

1.13 PF Status C Register

Address Offset: 0x0F

Field	Bits	Default	Type	Field Description
pf_status_c	[7:0]	0x0	RO	bit7: Commanded Permanent Fail bit6: Hardware Mux Permanent Fail bit5: Internal VSS Measurement Permanent Fail bit4: Internal Voltage Reference Permanent Fail bit3: Internal LFO Permanent Fail bit2:Instruction ROM Permanent Fail bit1:Data ROM Permanent Fail bit0:OTP Memory Permanent Fail 0 = Fault is not triggered. 1 = Fault is triggered.

1.14 PF Alert D Register

Address Offset: 0x10

Field	Bits	Default	Type	Field Description
pf_alert_d	[0:0]	0x0	RO	Top of Stack vs Cell Sum Permanent Fail 0 = Alert is not triggered. 1 = Alert is triggered.

1.15 PF Status D Register

Address Offset: 0x11

Field	Bits	Default	Type	Field Description
pf_status_d	[0:0]	0x0	RO	Top of Stack vs Cell Sum Permanent Fail 0 = Fault is not triggered. 1 = Fault is triggered.

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1.16 Battery Status Register

Address Offset: 0x12

Field	Bits	Default	Type	Field Description
sleep	[15:15]	0x0	RO	This bit indicates whether or not the device is presently in SLEEP mode. 0 = Device is not in SLEEP mode. 1 = Device is in SLEEP mode.
sd_cmd	[13:13]	0x0	RO	This bit is set when shutdown is pending because the command was received or the RST_SHUT pin was asserted for at least one second. 0 = Shutdown due to command or pin is not pending. 1 = Shutdown due to command or pin is pending.
pf	[12:12]	0x0	RO	This bit indicates whether or not an enabled Permanent Fail fault has triggered. 0 = No Permanent Fail fault has triggered. 1 = At least one Permanent Fail fault has triggered.
ss	[11:11]	0x0	RO	This bit indicates whether or not an enabled safety fault is triggered. 0 = No safety fault is triggered. 1 = At least one enabled safety fault is triggered.
fuse	[10:10]	0x0	RO	This bit reports the most recently observed state of the FUSE pin and is updated every second. 0 = FUSE pin was not asserted by device or secondary protector at last sample. 1 = FUSE pin was asserted by device or secondary protector at last sample.
sec	[9: 8]	0x0	RO	These bits indicate the present security state of the device. When in SEALED mode, device configuration may not be read or written and some commands are restricted. When in UNSEALED mode, device configuration may normally be read and may be written while in CONFIG_UPDATE mode. When in FULLACCESS mode, unrestricted read and write access is allowed and all commands are accepted. Even in FULLACCESS mode, changes to device configuration should only be changed while also in CONFIG_UPDATE mode. 0 = Device has not initialized yet. 1 = Device is in FULLACCESS mode. 2 = Device is in UNSEALED mode. 3 = Device is in SEALED mode

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Field	Bits	Default	Type	Field Description
otpb	[7:7]	0x0	RO	This bit indicates whether or not voltage and temperature conditions are valid for OTP programming. During normal operation, this bit will always be set if Manufacturing Status() [OTPW] is clear. When entering CONFIG_UPDATE mode, conditions will be checked and this bit will reflect whether or not programming is allowed (Manufacturing Status()[OTPW] does not apply in CONFIG_UPDATE mode). Once in CONFIG_UPDATE mode, this bit will not change state since no new measurements are being taken. 0 = OTP writes are allowed. 1 = Writes to OTP are blocked.
otpw	[6:6]	0x0	RO	This bit indicates whether or not some data is waiting to be written to OTP during normal operation. This can occur when, for example, configured to Permanent Fail information to OTP. This bit may remain set until conditions for OTP programming are met and all data is programmed. This bit is not set during OTP programming from CONFIG_UPDATE mode. 0 = No writes to OTP are pending. 1 = Writes to OTP are pending.
cow_chk	[5:5]	0x0	RO	This bit indicates while cell open-wire checks are occurring. When the feature is disabled, this bit will not set. When the feature is enabled, this bit will set periodically as the checks are performed. 0 = Device is not actively performing a cell open-wire check. 1 = Device is actively performing a cell open-wire check.
wd	[4:4]	0x0	RO	This bit indicates whether or not the previous device reset was caused by the internal watchdog timer. This is not related to the Host Watchdog protection. 0 = Previous reset was normal. 1 = Previous reset was caused by the watchdog timer.
por	[3:3]	0x0	RO	This bit is set when the device fully resets. It is cleared upon exit of CONFIG_UPDATE mode. It can be used by the host to determine if any RAM configuration changes were lost due to a reset. 0 = Full reset has not occurred since last exit of CONFIG_UPDATE mode. 1 = Full reset has occurred since last exit of CONFIG_UPDATE and reconfiguration of any RAM settings is required.
sleep_en	[2:2]	0x0	RO	This bit indicates whether or not SLEEP mode is allowed based on configuration and commands. The Settings:Configuration:Power Config[SLEEP] bit sets the default state of this bit. The host may send commands to enable or disable SLEEP mode based on system requirements. When this bit is set, the device may transition to SLEEP mode when other SLEEP criteria are met.

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Field	Bits	Default	Type	Field Description
				0 = SLEEP mode is disabled by the host. 1 = SLEEP mode is allowed when other SLEEP conditions are met.
pchg_mode	[1:1]	0x0	RO	This bit indicates whether or not the device is in PRECHARGE mode. In PRECHARGE mode, the PCHG FET is turned on instead of the CHG FET. 0 = Device is not in PRECHARGE mode. 1 = Device is in PRECHARGE mode.
cfgupdate	[0:0]	0x0	RO	This bit indicates whether or not the device is in CONFIG_UPDATE mode. It will be set after the SET_CFGUPDATE command is received and fully processed. Configuration settings may be changed only while this bit is set. 0 = Device is not in CONFIG_UPDATE mode. 1 = Device is in CONFIG_UPDATE mode.

1.17 Cell1 Voltage Register

Address Offset: 0x14

Field	Bits	Default	Type	Field Description
cell1_volt	[15:0]	0x0	RO	16-bit voltage on cell 1, the unit is mV.

1.18 Cell2 Voltage Register

Address Offset: 0x16

Field	Bits	Default	Type	Field Description
cell2_volt	[15:0]	0x0	RO	16-bit voltage on cell 2, the unit is mV.

1.19 Cell3 Voltage Register

Address Offset: 0x18

Field	Bits	Default	Type	Field Description
cell3_volt	[15:0]	0x0	RO	16-bit voltage on cell 3, the unit is mV.

1.20 Cell4 Voltage Register

Address Offset: 0x1A

Field	Bits	Default	Type	Field Description
cell4_volt	[15:0]	0x0	RO	16-bit voltage on cell 4, the unit is mV.

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1.21 Cell5 Voltage Register

Address Offset: 0x1C

Field	Bits	Default	Type	Field Description
cell5_volt	[15:0]	0x0	RO	16-bit voltage on cell 5, the unit is mV.

1.22 Cell6 Voltage Register

Address Offset: 0x1E

Field	Bits	Default	Type	Field Description
cell6_volt	[15:0]	0x0	RO	16-bit voltage on cell 6, the unit is mV.

1.23 Cell7 Voltage Register

Address Offset: 0x20

Field	Bits	Default	Type	Field Description
cell7_volt	[15:0]	0x0	RO	16-bit voltage on cell 7, the unit is mV.

1.24 Cell8 Voltage Register

Address Offset: 0x22

Field	Bits	Default	Type	Field Description
cell8_volt	[15:0]	0x0	RO	16-bit voltage on cell 8, the unit is mV.

1.25 Cell9 Voltage Register

Address Offset: 0x24

Field	Bits	Default	Type	Field Description
cell9_volt	[15:0]	0x0	RO	16-bit voltage on cell 9, the unit is mV.

1.26 RV Register

Address Offset: 0x26

Field	Bits	Default	Type	Field Description
reserved	[15:0]	0x0	RO	Reserved cell voltage, the unit is mV.

1.27 Cell10 Voltage Register

Address Offset: 0x28

Field	Bits	Default	Type	Field Description
cell10_volt	[15:0]	0x0	RO	16-bit voltage on cell 10, the unit is mV.

1.28 Cell11 Voltage Register

Address Offset: 0x2A

Field	Bits	Default	Type	Field Description
cell11_volt	[15:0]	0x0	RO	16-bit voltage on cell 11, the unit is mV.

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1.29 Cell12 Voltage Register

Address Offset: 0x2C

Field	Bits	Default	Type	Field Description
cell12_volt	[15:0]	0x0	RO	16-bit voltage on cell 12, the unit is mV.

1.30 Cell13 Voltage Register

Address Offset: 0x2E

Field	Bits	Default	Type	Field Description
Cell13_volt	[15:0]	0x0	RO	16-bit voltage on cell 13, the unit is mV.

1.31 Cell14 Voltage Register

Address Offset: 0x30

Field	Bits	Default	Type	Field Description
Cell14_volt	[15:0]	0x0	RO	16-bit voltage on cell 14, the unit is mV.

1.32 Cell15 Voltage Register

Address Offset: 0x32

Field	Bits	Default	Type	Field Description
Cell15_volt	[15:0]	0x0	RO	16-bit voltage on cell 15, the unit is mV.

1.33 Stack Voltage Register

Address Offset: 0x34

Field	Bits	Default	Type	Field Description
stack_volt	[15:0]	0x0	RO	16-bit voltage on top of stack, the unit is userV, default is 8mV

1.34 Pack Pin Voltage Register

Address Offset: 0x36

Field	Bits	Default	Type	Field Description
pack_pin_volt	[15:0]	0x0	RO	16-bit voltage on PACK pin, the unit is userV, default is 8mV

1.35 LD Pin Voltage Register

Address Offset: 0x38

Field	Bits	Default	Type	Field Description
ld_pin_volt	[15:0]	0x0	RO	16-bit voltage on LD pin, the unit is userV, default is 8mV

1.36 CC2 Current Register

Address Offset: 0x3A

Field	Bits	Default	Type	Field Description
cc2_curr	[15:0]	0x0	RO	16-bit CC2 current, the unit is userA. userA is defined in cc gain

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1.37 Cell16 Voltage Register

Address Offset: 0x3C

Field	Bits	Default	Type	Field Description
cell16_volt	[15: 0]	0x0	RO	16-bit voltage on cell 16, the unit is mV.

1.38 Alarm Status Register

Address Offset: 0x62, write 1 to clear

Field	Bits	Default	Type	Field Description
SSBC	[15:15]	0x0	LH	This bit is set when a bit is set in Safety Status B() or Safety Status C().
SSA	[14:14]	0x0	LH	This bit is set when a bit is set in Safety Status A().
PF	[13:13]	0x0	LH	This bit is set when an enabled Permanent Fail fault triggers.
MSK_SFALERT	[12:12]	0x0	LH	This bit is set when a safety alert is triggered that is also enabled in the corresponding Settings:Alarm:SF Alert Mask A, Settings:Alarm:SF Alert Mask B, or Settings:Alarm:SF Alert Mask C register.
MSK_PFALERT	[11:11]	0x0	LH	This bit is set when a Permanent Fail alert is triggered that is also enabled in the corresponding Settings:Alarm:PF Alert Mask A, Settings:Alarm:PF Alert Mask B, Settings:Alarm:PF Alert Mask C, or Settings:Alarm:PF Alert Mask D register.
INITSTART	[10:10]	0x0	LH	Initialization started (sets quickly after device powers up).
INITCOMP	[9:9]	0x0	LH	Initialization completed (sets after the device has powered and completed one measurement scan).
RSVD_0	[8: 8]	0x0	LH	RSVD_0
FULLSCAN	[7: 7]	0x0	LH	Full Voltage Scan Complete. The necessary multiple ADC scans have been completed to collect the full voltag measurement loop data (including cell voltages, pin or thermistor voltages, etc). This bit sets each time a full scan completes (when enabled).
XCHG	[6: 6]	0x0	LH	This bit is set when the CHG FET is off.
XDSG	[5: 5]	0x0	LH	This bit is set when the DSG FET is off.
SHUTV	[4: 4]	0x0	LH	Stack voltage is below Shutdown Stack Voltage.
FUSE	[3: 3]	0x0	LH	FUSE Pin Driven. FUSE pin is being driven by either the device or the secondary protector.
CB	[2: 2]	0x0	LH	This bit is set when cell balancing is active.
ADSCAN	[1: 1]	0x0	LH	Voltage ADC Scan Complete. A single ADC scan is complete (cell voltages are measured on each scan). This bit sets each time a scan completes (when enabled).
WAKE	[0: 0]	0x0	LH	This bit is set when the device is wakened from SLEEP mode.

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1.39 Alarm Raw Status Register

Address Offset: 0x64

Field	Bits	Default	Type	Field Description
SSBC	[15:15]	0x0	RO	This bit is set when a bit is set in Safety Status B() or Safety Status C().
SSA	[14:14]	0x0	RO	This bit is set when a bit is set in Safety Status A().
PF	[13:13]	0x0	RO	This bit is set when an enabled Permanent Fail fault triggers.
MSK_SFALERT	[12:12]	0x0	RO	This bit is set when a safety alert is triggered that is also enabled in the corresponding Settings:Alarm:SF Alert Mask A, Settings:Alarm:SF Alert Mask B, or Settings:Alarm:SF Alert Mask C register.
MSK_PFALERT	[11:11]	0x0	RO	This bit is set when a Permanent Fail alert is triggered that is also enabled in the corresponding Settings:Alarm:PF Alert Mask A, Settings:Alarm:PF Alert Mask B, Settings:Alarm:PF Alert Mask C, or Settings:Alarm:PF Alert Mask D register.
INITSTART	[10:10]	0x0	RO	Initialization started (sets quickly after device powers up).
INITCOMP	[9:9]	0x0	RO	Initialization completed (sets after the device has powered and completed one measurement scan).
RSVD_0	[8: 8]	0x0	RO	RSVD_0
FULLSCAN	[7: 7]	0x0	RO	Full Voltage Scan Complete. The necessary multiple ADC scans have been completed to collect the full voltag measurement loop data (including cell voltages, pin or thermistor voltages, etc). This bit sets each time a full scan completes (when enabled).
XCHG	[6: 6]	0x0	RO	This bit is set when the CHG FET is off.
XDSG	[5: 5]	0x0	RO	This bit is set when the DSG FET is off.
SHUTV	[4: 4]	0x0	RO	Stack voltage is below Shutdown Stack Voltage.
FUSE	[3: 3]	0x0	RO	FUSE Pin Driven. FUSE pin is being driven by either the device or the secondary protector.
CB	[2: 2]	0x0	RO	This bit is set when cell balancing is active.
ADSCAN	[1: 1]	0x0	RO	Voltage ADC Scan Complete. A single ADC scan is complete (cell voltages are measured on each scan). This bit sets each time a scan completes (when enabled).
WAKE	[0: 0]	0x0	RO	This bit is set when the device is wakened from SLEEP mode.

1.40 Alarm Enable Register

Address Offset: 0x66

Field	Bits	Default	Type	Field Description
SSBC	[15:15]	0x0	ROHW	When Alarm Raw Status()[SSBC] is active, setting this bit will cause Alarm Status()[SSBC] to be set and latched.
SSA	[14:14]	0x0	ROHW	When Alarm Raw Status()[SSA] is active, setting this bit will cause Alarm Status()[SSA] to be set and latched.
PF	[13:13]	0x0	ROHW	When Alarm Raw Status()[PF] is active, setting this bit will cause

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				Alarm Status()[PF] to be set and latched.
MSK_SFALERT	[12:12]	0x0	ROHW	When Alarm Raw Status()[MSK_SFALERT] is active, setting this bit will cause Alarm Status()[MSK_SFALERT] to be set and latched.
MSK_PFALERT	[11:11]	0x0	ROHW	When Alarm Raw Status()[MSK_PFALERT] is active, setting this bit will cause Alarm Status()[MSK_PFALERT] to be set and latched.
INITSTART	[10:10]	0x0	ROHW	When Alarm Raw Status()[INITSTART] is active, setting this bit will cause Alarm Status()[INITSTART] to be set and latched.
INITCOMP	[9:9]	0x0	ROHW	When Alarm Raw Status()[INITCOMP] is active, setting this bit will cause Alarm Status()[INITCOMP] to be set and latched.
RSVD_0	[8: 8]	0x0	ROHW	RSVD_0
FULLSCAN	[7: 7]	0x0	ROHW	When Alarm Raw Status()[FULLSCAN] is active, setting this bit will cause Alarm Status()[FULLSCAN] to be set and latched.
XCHG	[6: 6]	0x0	ROHW	When Alarm Raw Status()[XCHG] is active, setting this bit will cause Alarm Status()[XCHG] to be set and latched.
XDSG	[5: 5]	0x0	ROHW	When Alarm Raw Status()[XDSG] is active, setting this bit will cause Alarm Status()[XDSG] to be set and latched
SHUTV	[4: 4]	0x0	ROHW	When Alarm Raw Status()[SHUTV] is active, setting this bit will cause Alarm Status()[SHUTV] to be set and latched.
FUSE	[3: 3]	0x0	ROHW	When Alarm Raw Status()[FUSE] is active, setting this bit will cause Alarm Status()[FUSE] to be set and latched.
CB	[2: 2]	0x0	ROHW	When Alarm Raw Status()[CB] is active, setting this bit will cause Alarm Status()[CB] to be set and latched.
ADSCAN	[1: 1]	0x0	ROHW	When Alarm Raw Status()[ADSCAN] is active, setting this bit will cause Alarm Status()[ADSCAN] to be set and latched.
WAKE	[0: 0]	0x0	ROHW	When Alarm Raw Status()[WAKE] is active, setting this bit will cause Alarm Status()[WAKE] to be set and latched.

1.41 INT Temperature Register

Address Offset: 0x68

Field	Bits	Default	Type	Field Description
int_temp	[15:0]	0x0	RO	This is the most recent measured internal die temperature, the unit is 0.1℃.

1.42 CFEOFF Temperature Register

Address Offset: 0x6A

Field	Bits	Default	Type	Field Description
cfetoff_temp	[15:0]	0x0	RO	When the CFETOFF pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the cfetoff pin in mV.

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1.43 DFEOFF Temperature Register

Address Offset: 0x6C

Field	Bits	Default	Type	Field Description
dfetoff_temp	[15:0]	0x0	RO	When the DFETOFF pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the dfetoff pin in mV.

1.44 Alert Temperature Register

Address Offset: 0x6E

Field	Bits	Default	Type	Field Description
alert_temp	[15:0]	0x0	RO	When the ALERT pin is configured as a thermistor input, this reports its most recent temperature measurement, the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the Alert pin in mV.

1.45 TS1 Temperature Register

Address Offset: 0x70

Field	Bits	Default	Type	Field Description
ts1_temp	[15:0]	0x0	RO	When the TS1 pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the TS1 pin in mV.

1.46 TS2 Temperature Register

Address Offset: 0x72

Field	Bits	Default	Type	Field Description
ts2_temp	[15:0]	0x0	RO	When the TS2 pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the TS2 pin in mV.

1.47 TS3 Temperature Register

Address Offset: 0x74

Field	Bits	Default	Type	Field Description
ts3_temp	[15:0]	0x0	RO	When the TS3 pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the TS3 pin in mV.

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1.48 SPI MOSI Temperature Register

Address Offset: 0x76

Field	Bits	Default	Type	Field Description
spi_mosi_temp	[15:0]	0x0	RO	When the SPI_MOSI pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the SPI MOSI pin in mV.

1.49 DCHG Temperature Register

Address Offset: 0x78

Field	Bits	Default	Type	Field Description
dchg_temp	[15:0]	0x0	RO	When the DCHG pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the DCHG pin in mV.

1.50 DDSG Temperature Register

Address Offset: 0x7A

Field	Bits	Default	Type	Field Description
ddsg_temp	[15:0]	0x0	RO	When the DDSG pin is configured as a thermistor input, this reports its most recent temperature measurement，the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the DDSG pin in mV.

1.51 TS0 Temperature Register

Address Offset: 0x7C

Field	Bits	Default	Type	Field Description
ts0_temp	[15:0]	0x0	RO	When the TS0 pin is configured as a thermistor input, this reports its most recent temperature measurement, the unit is 0.1℃. When configured as ADCIN, this instead reports the measured voltage at the TS0 pin in mV.

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1.52 FET Status Register

Address Offset: 0x7F

Field	Bits	Default	Type	Field Description
alrt_pin	[6:6]	0x0	RO	Indicates the status of the ALERT pin. 0 = The ALERT pin is not asserted. 1 = The ALERT pin is asserted.
ddsg_pin	[5:5]	0x0	RO	Indicates the status of the DDSG pin. 0 = The DDSG pin is not asserted. 1 = The DDSG pin is asserted.
dchg_pin	[4:4]	0x0	RO	Indicates the status of the DCHG pin. 0 = The DCHG pin is not asserted. 1 = The DCHG pin is asserted.
pdsg_fet	[3:3]	0x0	RO	Indicates the status of the PDSG FET. 0 = The PDSG FET is off. 1 = The PDSG FET is on
dsg_fet	[2:2]	0x0	RO	Indicates the status of the DSG FET. 0 = The DSG FET is off. 1 = The DSG FET is on.
pchg_fet	[1:1]	0x0	RO	Indicates the status of the PCHG FET. 0 = The PCHG FET is off. 1 = The PCHG FET is on.
chg_fet	[0:0]	0x0	RO	Indicates the status of the CHG FET. 0 = The CHG FET is off. 1 = The CHG FET is on.

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2 Command-Only Subcommands

Base Address: 0x0300

2.1 Exit Deepsleep Register

Address Offset: 0x00

Field	Bits	Default	Type	Field Description
exit_deepsleep	[7:0]	0x0	WO	sub cmd only

2.2 Deepsleep Register

Address Offset: 0x04

Field	Bits	Default	Type	Field Description
deepsleep	[7:0]	0x0	WO	sub cmd only, this command needs to be sent twice continuously within 4 seconds.

2.3 Shutdown Register

Address Offset: 0x08

Field	Bits	Default	Type	Field Description
shutdown	[7:0]	0x0	WO	sub cmd only, this command needs to be sent twice continuously within 4 seconds.

2.4 Reset Register

Address Offset: 0x0C

Field	Bits	Default	Type	Field Description
sw_reset	[7:0]	0x0	WO	sub cmd only, this command will put the AFE into shutdown mode, and its use is not recommended. Please use the shutdown command to control it.

2.5 PDSGTest Register

Address Offset: 0x10

Field	Bits	Default	Type	Field Description
pdsgtest	[7:0]	0x0	WO	sub cmd only, this cmd a toggle control.

2.6 Fuse Toggle Register

Address Offset: 0x14

Field	Bits	Default	Type	Field Description
fuse_toggle	[7:0]	0x0	WO	sub cmd only

2.7 PCHGTest Register

Address Offset: 0x18

Field	Bits	Default	Type	Field Description
pchgtest	[7:0]	0x0	WO	sub cmd only, this cmd a toggle control.

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2.8 CHGTest Register

Address Offset: 0x1C

Field	Bits	Default	Type	Field Description
chgtest	[7:0]	0x0	WO	sub cmd only, this cmd a toggle control.

2.9 DSGTest Register

Address Offset: 0x20

Field	Bits	Default	Type	Field Description
dsgtest	[7:0]	0x0	WO	sub cmd only, this cmd a toggle control.

2.10 FET Enable Register

Address Offset: 0x24

Field	Bits	Default	Type	Field Description
fet_enable	[7:0]	0x0	WO	sub cmd only, this is a toggle signal

2.11 PF Enable Register

Address Offset: 0x28

Field	Bits	Default	Type	Field Description
pf_enable	[7:0]	0x0	WO	sub cmd only, this is a toggle signal.

2.12 Seal Register

Address Offset: 0x2C

Field	Bits	Default	Type	Field Description
seal	[7:0]	0x0	WO	sub cmd only

2.13 Reset PassQ Register

Address Offset: 0x30

Field	Bits	Default	Type	Field Description
reset_passq	[7:0]	0x0	WO	sub cmd only

2.14 PTO Recover Register

Address Offset: 0x34

Field	Bits	Default	Type	Field Description
pto_recover	[7:0]	0x0	WO	sub cmd only

2.15 SET Cfgupdate Register

Address Offset: 0x38

Field	Bits	Default	Type	Field Description
set_cfgupdate	[7:0]	0x0	WO	sub cmd only

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2.16 Exit Cfgupdate Register

Address Offset: 0x3C

Field	Bits	Default	Type	Field Description
exit_cfgupdate	[7:0]	0x0	WO	sub cmd only

2.17 DSG PDSG Off Register

Address Offset: 0x40

Field	Bits	Default	Type	Field Description
dsg_pdsg_off	[7:0]	0x0	WO	sub cmd only

2.18 CHG PCHG Off Register

Address Offset: 0x44

Field	Bits	Default	Type	Field Description
chg_pchg_off	[7:0]	0x0	WO	sub cmd only

2.19 All FETs Off Register

Address Offset: 0x48

Field	Bits	Default	Type	Field Description
all_fets_off	[7:0]	0x0	WO	sub cmd only

2.20 All FETs On Register

Address Offset: 0x4C

Field	Bits	Default	Type	Field Description
all_fets_on	[7:0]	0x0	WO	sub cmd only

2.21 Sleep Enable Register

Address Offset: 0x50

Field	Bits	Default	Type	Field Description
sleep_enable	[7:0]	0x0	WO	sub cmd only

2.22 Sleep Disable Register

Address Offset: 0x54

Field	Bits	Default	Type	Field Description
sleep_disable	[7:0]	0x0	WO	sub cmd only

2.23 OCDL Recover Register

Address Offset: 0x58

Field	Bits	Default	Type	Field Description
ocdl_recover	[7:0]	0x0	WO	sub cmd only

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2.24 SCDL Recover Register

Address Offset: 0x5C

Field	Bits	Default	Type	Field Description
scdl_recover	[7:0]	0x0	WO	sub cmd only

2.25 Load Detect Restart Register

Address Offset: 0x60

Field	Bits	Default	Type	Field Description
load_detect_restart	[7:0]	0x0	WO	sub cmd only

2.26 Load Detect On Register

Address Offset: 0x64

Field	Bits	Default	Type	Field Description
load_detect_on	[7:0]	0x0	WO	sub cmd only

2.27 Load Detect Off Register

Address Offset: 0x68

Field	Bits	Default	Type	Field Description
load_detect_off	[7:0]	0x0	WO	sub cmd only

2.28 CFETOFF Lo Register

Address Offset: 0x6C

Field	Bits	Default	Type	Field Description
cfetoff_lo	[7:0]	0x0	WO	sub cmd only

2.29 DFETOFF Lo Register

Address Offset: 0x70

Field	Bits	Default	Type	Field Description
dfetoff_lo	[7:0]	0x0	WO	sub cmd only

2.30 Alert Lo Register

Address Offset: 0x74

Field	Bits	Default	Type	Field Description
alert_lo	[7:0]	0x0	WO	sub cmd only

2.31 SPI MOSI Lo Register

Address Offset: 0x78

Field	Bits	Default	Type	Field Description
spi_mosi_lo	[7:0]	0x0	WO	sub cmd only

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2.32 DCHG Lo Register

Address Offset: 0x7C

Field	Bits	Default	Type	Field Description
dchg_lo	[7:0]	0x0	WO	sub cmd only

2.33 DDSG Lo Register

Address Offset: 0x80

Field	Bits	Default	Type	Field Description
ddsg_lo	[7:0]	0x0	WO	sub cmd only

2.34 CFETOff Hi Register

Address Offset: 0x84

Field	Bits	Default	Type	Field Description
cfetoff_hi	[7:0]	0x0	WO	sub cmd only

2.35 DFETOff Hi Register

Address Offset: 0x88

Field	Bits	Default	Type	Field Description
dfetoff_hi	[7:0]	0x0	WO	sub cmd only

2.36 Alert Hi Register

Address Offset: 0x8C

Field	Bits	Default	Type	Field Description
alert_hi	[7:0]	0x0	WO	sub cmd only

2.37 SPI MOSI Hi Register

Address Offset: 0x90

Field	Bits	Default	Type	Field Description
spi_mosi_hi	[7:0]	0x0	WO	sub cmd only

2.38 DCHG Hi Register

Address Offset: 0x94

Field	Bits	Default	Type	Field Description
dchg_hi	[7:0]	0x0	WO	sub cmd only

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2.39 DDSG Hi Register

Address Offset: 0x98

Field	Bits	Default	Type	Field Description
ddsg_hi	[7:0]	0x0	WO	sub cmd only

2.40 PF Force A Register

Address Offset: 0x9C

Field	Bits	Default	Type	Field Description
pf_force_a	[7:0]	0x0	WO	sub cmd only

2.41 PF Force B Register

Address Offset: 0xA0

Field	Bits	Default	Type	Field Description
pf_force_b	[7:0]	0x0	WO	sub cmd only

2.42 SWAP Comm Mode Register

Address Offset: 0xA4

Field	Bits	Default	Type	Field Description
swap_comm_mode	[7:0]	0x0	WO	sub cmd only

2.43 SWAP to I2C Register

Address Offset: 0xA8

Field	Bits	Default	Type	Field Description
swap_to_i2c	[7:0]	0x0	WO	sub cmd only

2.44 SWAP to SPI Register

Address Offset: 0xAC

Field	Bits	Default	Type	Field Description
swap_to_spi	[7:0]	0x0	WO	sub cmd only

2.45 Sleep Register

Address Offset: 0xB4

Field	Bits	Default	Type	Field Description
sleep	[7:0]	0x0	WO	sub cmd only

2.46 SPICS Lo Register

Address Offset: 0xC0

Field	Bits	Default	Type	Field Description
spics_lo	[7:0]	0x0	WO	sub cmd only

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2.47 SPICS Hi Register

Address Offset: 0xC4

Field	Bits	Default	Type	Field Description
spics_hi	[7:0]	0x0	WO	sub cmd only

2.48 Normal INTR Clr Register

Address Offset: 0xC8

Field	Bits	Default	Type	Field Description
normal_intr_clr	[7:0]	0x0	WO	sub cmd only

2.49 Chip Reset Register

Address Offset: 0xCC

Field	Bits	Default	Type	Field Description
chip_reset	[7:0]	0x0	WO	sub cmd only, this command will reset the chip.

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3 Subcommands with Data

Base Address: 0x0000

3.1 Device Number Register

Address Offset: 0x00

Field	Bits	Default	Type	Field Description
device_num	[15:0]	0x0	RO	Reports the device number that identifies the product. The data is returned in little-endian format.

3.2 HW Version Register

Address Offset: 0x0C

Field	Bits	Default	Type	Field Description
hardware_version	[15:0]	0x0	RO	Reports the device hardware version number.

3.3 Test State 0 Register

Address Offset: 0x10

Field	Bits	Default	Type	Field Description
sysc_state	[15:0]	0x0	RO	sysc test state

3.4 Test State 1 Register

Address Offset: 0x12

Field	Bits	Default	Type	Field Description
otp_state	[7:0]	0x0	RO	otp test state

3.5 Test State 2 Register

Address Offset: 0x13

Field	Bits	Default	Type	Field Description
adc_state	[7:0]	0x0	RO	adc test state

3.6 Test State 3 Register

Address Offset: 0x14

Field	Bits	Default	Type	Field Description
pt_state	[7:0]	0x0	RO	pt test state

3.7 Test State 4 Register

Address Offset: 0x15

Field	Bits	Default	Type	Field Description
cb_state	[7:0]	0x0	RO	cb test state

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3.8 Security KeyS0 Register

Address Offset: 0x20

Field	Bits	Default	Type	Field Description
unseal_key_step1	[15:0]	0x5a5a	ROHW	This is the first word of the security key that must be sent to transition from SEALED to UNSEALED mode.

3.9 Security KeyS1 Register

Address Offset: 0x22

Field	Bits	Default	Type	Field Description
unseal_key_step2	[15:0]	0x5a5a	ROHW	This is the second word of the security key that must be sent to transition from SEALED to UNSEALED mode. It must be sent within 5 seconds of the first word of the key and with no other commands in between.

3.10 Security KeyS2 Register

Address Offset: 0x24

Field	Bits	Default	Type	Field Description
full_acc_key_step1	[15:0]	0xa5a5	ROHW	This is the first word of the security key that must be sent to transition from UNSEALED to FULLACCESS mode.

3.11 Security KeyS3 Register

Address Offset: 0x26

Field	Bits	Default	Type	Field Description
full_acc_key_step2	[15:0]	0xa5a5	ROHW	This is the second word of the security key that must be sent to transition from UNSEALED to FULLACCESS mode. It must be sent within 5 seconds of the first word of the key and with no other commands in between.

3.12 Saved PF Status0 Register

Address Offset: 0x28

Field	Bits	Default	Type	Field Description
pf_statusa	[7:0]	0x0	RO	Saved Permanent Failure Status A Bit0: SUV Bit1: SOV, Bit2: SOCC, Bit3: SOCD, Bit4: SOT, Bit5: RSVD_0 Bit6: SOTF, Bit7: CUDEP,

3.13 Saved PF Status1 Register

Address Offset: 0x29

Field	Bits	Default	Type	Field Description
pf_statusb	[7:0]	0x0	RO	Saved Permanent Failure Status B Bit0: CFETF Bit1: DFETF Bit2: 2LVL Bit3: VIMR Bit4: VIMA Bit5: RSVD_0 Bit6: RSVD_0 Bit7: SCDL

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3.14 Saved PF Status2 Register

Address Offset: 0x2A

Field	Bits	Default	Type	Field Description
pf_statusc	[7:0]	0x0	RO	Saved Permanent Failure Status C Bit0: OTPF Bit1: DRMF Bit2: IRMF Bit3: LFOF Bit4: VREF Bit5: VSSF Bit6: HWMX Bit7: RSVD_0

3.15 Saved PF Status3 Register

Address Offset: 0x2B

Field	Bits	Default	Type	Field Description
pf_statusd	[7:0]	0x0	RO	Saved Permanent Failure Status D Bit1 - 7: RSVD_0 Bit0: TOSF

3.16 Saved PF Status4 Register

Address Offset: 0x2C

Field	Bits	Default	Type	Field Description
fuse_flag	[7:0]	0x0	RO	This is used to track whether or not the fuse has already been blown. This byte is normally zero but is set to 0x72 after the fuse is blown.

3.17 Manufacturing Status Register

Address Offset: 0x30

Field	Bits	Default	Type	Field Description
RSVD_0	[15]	0x0	RO	reserved
RSVD_0	[14]	0x0	RO	reserved
RSVD_0	[13]	0x0	RO	reserved
RSVD_0	[12]	0x0	RO	reserved
RSVD_0	[11]	0x0	RO	reserved
RSVD_0	[10]	0x0	RO	reserved
RSVD_0	[9]	0x0	RO	reserved
RSVD_0	[8]	0x0	RO	reserved
OTPW_EN	[7]	0x0	RO	allows/disallows writes to OTP during normal operation
PF_EN	[6]	0x0	RO	enables/disables Permanent Failure checks
PDSG_TEST	[5]	0x0	RO	indicates whether PDSG FET is enabled in FET Test Mode
FET_EN	[4]	0x0	RO	controls switching between FET Test Mode and normal FET control
RSVD_0	[3]	0x0	RO	reserved
DSG_TEST	[2]	0x0	RO	indicates whether DSG FET is enabled in FET Test Mode
CHG_TEST	[1]	0x0	RO	indicates whether CHG FET is enabled in FET Test Mode
PCHG_TEST	[0]	0x0	RO	indicates whether PCHG FET is enabled in FET Test Mode

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3.18 DAStatus1_0 Register

Address Offset: 0x80

Field	Bits	Default	Type	Field Description
cell1_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.19 DAStatus1_1 Register

Address Offset: 0x84

Field	Bits	Default	Type	Field Description
cell1_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.20 DAStatus1_2 Register

Address Offset: 0x88

Field	Bits	Default	Type	Field Description
cell2_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.21 DAStatus1_3 Register

Address Offset: 0x8C

Field	Bits	Default	Type	Field Description
cell2_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.22 DAStatus1_4 Register

Address Offset: 0x90

Field	Bits	Default	Type	Field Description
cell3_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.23 DAStatus1_5 Register

Address Offset: 0x94

Field	Bits	Default	Type	Field Description
cell3_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.24 DAStatus1_6 Register

Address Offset: 0x98

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Field	Bits	Default	Type	Field Description
cell4_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.25 DAStatus1_7 Register

Address Offset: 0x9C

Field	Bits	Default	Type	Field Description
cell4_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit

3.26 DAStatus2_0 Register

Address Offset: 0xA0

Field	Bits	Default	Type	Field Description
cell5_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.27 DAStatus2_1 Register

Address Offset: 0xA4

Field	Bits	Default	Type	Field Description
cell5_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.28 DAStatus2_2 Register

Address Offset: 0xA8

Field	Bits	Default	Type	Field Description
cell6_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.29 DAStatus2_3 Register

Address Offset: 0xAC

Field	Bits	Default	Type	Field Description
cell6_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.30 DAStatus2_4 Register

Address Offset: 0xB0

Field	Bits	Default	Type	Field Description
cell7_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

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3.31 DAStatus2_5 Register

Address Offset: 0xB4

Field	Bits	Default	Type	Field Description
cell7_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.32 DAStatus2_6 Register

Address Offset: 0xB8

Field	Bits	Default	Type	Field Description
cell8_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.33 DAStatus2_7 Register

Address Offset: 0xBC

Field	Bits	Default	Type	Field Description
cell8_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.34 DAStatus3_0 Register

Address Offset: 0xC0

Field	Bits	Default	Type	Field Description
cell9_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.35 DAStatus3_1 Register

Address Offset: 0xC4

Field	Bits	Default	Type	Field Description
cell9_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit

3.36 DAStatus3_2 Register

Address Offset: 0xC8

Field	Bits	Default	Type	Field Description
reserved	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.37 DAStatus3_3 Register

Address Offset: 0xCC

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Field	Bits	Default	Type	Field Description
reserved	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.38 DAStatus3_4 Register

Address Offset: 0xD0

Field	Bits	Default	Type	Field Description
cell10_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.39 DAStatus3_5 Register

Address Offset: 0xD4

Field	Bits	Default	Type	Field Description
cell10_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.40 DAStatus3_6 Register

Address Offset: 0xD8

Field	Bits	Default	Type	Field Description
cell11_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.41 DAStatus3_7 Register

Address Offset: 0xDC

Field	Bits	Default	Type	Field Description
cell11_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.42 DAStatus4_0 Register

Address Offset: 0xE0

Field	Bits	Default	Type	Field Description
cell12_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.43 DAStatus4_1 Register

Address Offset: 0xE4

Field	Bits	Default	Type	Field Description
cell12_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. the most significant bit.

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3.44 DAStatus4_2 Register

Address Offset: 0xE8

Field	Bits	Default	Type	Field Description
cell13_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.45 DAStatus4_3 Register

Address Offset: 0xEC

Field	Bits	Default	Type	Field Description
cell13_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.46 DAStatus4_4 Register

Address Offset: 0xF0

Field	Bits	Default	Type	Field Description
cell14_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.47 DAStatus4_5 Register

Address Offset: 0xF4

Field	Bits	Default	Type	Field Description
cell14_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.48 DAStatus4_6 Register

Address Offset: 0xF8

Field	Bits	Default	Type	Field Description
cell15_volt_counts	[23:0]	0x0	RO	32-bit ADC counts for cell voltage measurement, the unit is 0.722uV.

3.49 DAStatus4_7 Register

Address Offset: 0xFC

Field	Bits	Default	Type	Field Description
cell15_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV. The most significant bit is the sign bit.

3.50 DAStatus5_0 Register

Address Offset: 0x100

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Field	Bits	Default	Type	Field Description
vreg18	[15:0]	0x0	RO	16-bit ADC count of the VREG18 measurement, which is used as a proxy to check the coulomb counter reference voltage. Provide a ratio information between a reference value and another reference value, measured for diagnostic purposes, which remains approximately constant.

3.51 DAStatus5_1 Register

Address Offset: 0x102

Field	Bits	Default	Type	Field Description
vss	[15:0]	0x0	RO	16-bit ADC count of the VSS pin. This is measured for diagnostic purposes to ensure the ADC input mux is working properly.

3.52 DAStatus5_2 Register

Address Offset: 0x104

Field	Bits	Default	Type	Field Description
max_cell_volt	[15:0]	0x0	RO	Maximum Cell Voltage, the unit is mV.

3.53 DAStatus5_3 Register

Address Offset: 0x106

Field	Bits	Default	Type	Field Description
min_cell_volt	[15:0]	0x0	RO	Minimum Cell Voltage, the unit is mV.

3.54 DAStatus5_4 Register

Address Offset: 0x108

Field	Bits	Default	Type	Field Description
battery_volt_sum	[15:0]	0x0	RO	Sum of cell voltages (including interconnects). This can be compared to the Stack Voltage() for diagnostic purposes. Note, however, that these measurements are taken at different times which may cause variation , the unit is userV, default is 8mV.

3.55 DAStatus5_5 Register

Address Offset: 0x10A

Field	Bits	Default	Type	Field Description
cell_temp	[15:0]	0x0	RO	Reports the cell temperature being used for features depending on a single temperature threshold. Note many features (such as protections) use minimum or maximum cell temperature instead. The unit is 0.1℃.

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3.56 DAStatus5_6 Register

Address Offset: 0x10C

Field	Bits	Default	Type	Field Description
fet_temp	[15:0]	0x0	RO	Reports the FET temperature given by the maximum of all measured FET temperatures. The unit is 0.1℃.

3.57 DAStatus5_7 Register

Address Offset: 0x10E

Field	Bits	Default	Type	Field Description
max_cell_temp	[15:0]	0x0	RO	Reports the maximum of all measured cell temperatures. The unit is 0.1℃.

3.58 DAStatus5_8 Register

Address Offset: 0x110

Field	Bits	Default	Type	Field Description
min_cell_temp	[15:0]	0x0	RO	Reports the minimum of all measured cell temperatures. The unit is 0.1℃.

3.59 DAStatus5_10 Register

Address Offset: 0x114

Field	Bits	Default	Type	Field Description
cc3_curr	[15:0]	0x0	RO	Reports the CC3 Current, which is obtained by averaging a configurable number of CC2 current measurements. The unit is userA., userA is user-defined in cc gain. The most significant bit is the sign bit, the range of a 16-bit value is from -32768 to +32767.

3.60 DAStatus5_11 Register

Address Offset: 0x116

Field	Bits	Default	Type	Field Description
cc1_curr	[15:0]	0x0	RO	Reports the CC1 current, which is updated every 100 ms in NORMAL mode. In SLEEP mode, this is updated every 4 seconds beginning one second after voltage measurements end. See the documentation on power modes for further details. The unit is userA, userA is user-defined in cc gain. The most significant bit is the sign bit, the range of a 16-bit value is from -32768 to +32767.

3.61 DAStatus5_12 Register

Address Offset: 0x118

Field	Bits	Default	Type	Field Description
cc2_counts	[31:0]	0x0	RO	Raw 32-bit count value for the latest CC2 measurement, the unit is 28.896nV. The most significant bit is the sign bit.

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3.62 DAStatus5_13 Register

Address Offset: 0x11C

Field	Bits	Default	Type	Field Description
cc3_counts	[31:0]	0x0	RO	For the 32-bit programmable CC2 average current, the high 16 bits represent the integer part and the low 8 bits represent the decimal part. The high 16 bits are taken as CC3 current.

3.63 DAStatus6_0 Register

Address Offset: 0x120

Field	Bits	Default	Type	Field Description
accum_charge	[31:0]	0x0	RO	Reports the integer portion of accumulated passed charge in userAmp-hours.

3.64 DAStatus6_1 Register

Address Offset: 0x124

Field	Bits	Default	Type	Field Description
accum_charge_fraction	[31:0]	0x0	RO	Reports the fractional portion of accumulated passed charge. This is initialized to 0.5 userAh to facilitate appropriate rounding of the integer portion.

3.65 DAStatus6_2 Register

Address Offset: 0x128

Field	Bits	Default	Type	Field Description
accum_time	[31:0]	0x0	RO	Reports the number of seconds over which passed charge has been integrated. The unit is S.

3.66 DAStatus6_3 Register

Address Offset: 0x12C

Field	Bits	Default	Type	Field Description
cfetoff_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the CFETOFF pin. The unit is 0.358uV.

3.67 DAStatus6_4 Register

Address Offset: 0x130

Field	Bits	Default	Type	Field Description
dfetoff_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the DFETOFF pin. The unit is 0.358uV.

3.68 DAStatus6_5 Register

Address Offset: 0x134

Field	Bits	Default	Type	Field Description

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alert_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the ALERT pin. The unit is 0.358uV.

3.69 DAStatus6_6 Register

Address Offset: 0x138

Field	Bits	Default	Type	Field Description
ts1_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the TS1 pin. When configured for thermistor measurement, the unit is 0.358uV.

3.70 DAStatus6_7 Register

Address Offset: 0x13C

Field	Bits	Default	Type	Field Description
ts2_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the TS2 pin. The unit is 0.358uV.

3.71 DAStatus7_0 Register

Address Offset: 0x140

Field	Bits	Default	Type	Field Description
ts3_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the TS3 pin. When configured for thermistor measurement, the unit is 0.358uV.

3.72 DAStatus7_2 Register

Address Offset: 0x144

Field	Bits	Default	Type	Field Description
spi_mosi_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the SPI_MOSI pin. The unit is 0.358uV.

DAStatus7_3 Register

Address Offset: 0x148

Field	Bits	Default	Type	Field Description
dchg_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the DCHG pin. The unit is 0.358uV.

3.73 DAStatus7_4 Register

Address Offset: 0x14C

Field	Bits	Default	Type	Field Description
ddsg_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the DDSG pin. The unit is 0.358uV.

3.74 DAStatus7_5 Register

Address Offset: 0x150

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Field	Bits	Default	Type	Field Description
cell16_curr_counts	[15:0]	0x0	RO	32-bit ADC counts for current measurement taken during the cell voltage measurement. the unit is 28.896nV.

3.75 DAStatus7_6 Register

Address Offset: 0x154

Field	Bits	Default	Type	Field Description
cell16_volt_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the Cell16. the unit is 0.722uV.

3.76 DAStatus7_7 Register

Address Offset: 0x158

Field	Bits	Default	Type	Field Description
cell16_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.77 DAStatus7_8 Register

Address Offset: 0x15A

Field	Bits	Default	Type	Field Description
cell16_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.78 DAStatus7_9 Register

Address Offset: 0x15C

Field	Bits	Default	Type	Field Description
Cell16_cb_time	[31:0]	0x0	RO	Cell 16 Total Balancing Time, The unit is S.

3.79 CUV Snapshot 0 Register

Address Offset: 0x180

Field	Bits	Default	Type	Field Description
cell1_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.80 CUV Snapshot 1 Register

Address Offset: 0x182

Field	Bits	Default	Type	Field Description
cell2_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.81 CUV Snapshot 2 Register

Address Offset: 0x184

Field	Bits	Default	Type	Field Description

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cell3_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.82 CUV Snapshot 3 Register

Address Offset: 0x186

Field	Bits	Default	Type	Field Description
cell4_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.83 CUV Snapshot 4 Register

Address Offset: 0x188

Field	Bits	Default	Type	Field Description
cell5_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.84 CUV Snapshot 5 Register

Address Offset: 0x18A

Field	Bits	Default	Type	Field Description
cell6_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.85 CUV Snapshot 6 Register

Address Offset: 0x18C

Field	Bits	Default	Type	Field Description
cell7_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.86 CUV Snapshot 7 Register

Address Offset: 0x18E

Field	Bits	Default	Type	Field Description
cell8_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.87 CUV Snapshot 8 Register

Address Offset: 0x190

Field	Bits	Default	Type	Field Description
cell9_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.88 CUV Snapshot 9 Register

Address Offset: 0x192

Field	Bits	Default	Type	Field Description

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reserved	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.89 CUV Snapshot 10 Register

Address Offset: 0x194

Field	Bits	Default	Type	Field Description
cell10_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.90 CUV Snapshot 11 Register

Address Offset: 0x196

Field	Bits	Default	Type	Field Description
cell11_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.91 CUV Snapshot 12 Register

Address Offset: 0x198

Field	Bits	Default	Type	Field Description
cell12_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.92 CUV Snapshot 13 Register

Address Offset: 0x19A

Field	Bits	Default	Type	Field Description
cell13_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.93 CUV Snapshot 14 Register

Address Offset: 0x19C

Field	Bits	Default	Type	Field Description
cell14_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.94 CUV Snapshot 15 Register

Address Offset: 0x19E

Field	Bits	Default	Type	Field Description
cell15_volt_cuv	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest CUV event.

3.95 COV Snapshot 0 Register

Address Offset: 0x1A0

ield	Bits	Default	Type	Field Description

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cell1_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.96 COV Snapshot 1 Register

Address Offset: 0x1A2

Field	Bits	Default	Type	Field Description
cell2_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.97 COV Snapshot 2 Register

Address Offset: 0x1A4

Field	Bits	Default	Type	Field Description
cell3_volt_cov	[15: 0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.98 COV Snapshot 3 Register

Address Offset: 0x1A6

Field	Bits	Default	Type	Field Description
cell4_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.99 COV Snapshot 4 Register

Address Offset: 0x1A8

Field	Bits	Default	Type	Field Description
cell5_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.100 COV Snapshot 5 Register

Address Offset: 0x1AA

Field	Bits	Default	Type	Field Description
cell6_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.101 COV Snapshot 6 Register

Address Offset: 0x1AC

Field	Bits	Default	Type	Field Description
cell7_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.102 COV Snapshot 7 Register

Address Offset: 0x1AE

Field	Bits	Default	Type	Field Description

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cell8_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.103 COV Snapshot 8 Register

Address Offset: 0x1B0

Field	Bits	Default	Type	Field Description
cell9_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.104 COV Snapshot 9 Register

Address Offset: 0x1B2

Field	Bits	Default	Type	Field Description
reserved	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.105 COV Snapshot 10 Register

Address Offset: 0x1B4

Field	Bits	Default	Type	Field Description
cell10_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.106 COV Snapshot 11 Register

Address Offset: 0x1B6

Field	Bits	Default	Type	Field Description
cell11_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.107 COV Snapshot 12 Register

Address Offset: 0x1B8

Field	Bits	Default	Type	Field Description
cell12_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.108 COV Snapshot 13 Register

Address Offset: 0x1BA

Field	Bits	Default	Type	Field Description
cell13_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.109 COV Snapshot 14 Register

Address Offset: 0x1BC

Field	Bits	Default	Type	Field Description

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cell14_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.110 COV Snapshot 15 Register

Address Offset: 0x1BE

Field	Bits	Default	Type	Field Description
cell15_volt_cov	[15:0]	0x0	RO	Records the cell voltage measurement made just after the latest COV event.

3.111 CB Active Cells Register

Address Offset: 0x200, Cell Balancing Active Cells

Field	Bits	Default	Type	Field Description
cb_active_cell16	[16]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell15	[15]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell14	[14]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell13	[13]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell12	[12]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell11	[11]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell10	[10]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
Reserved	[9]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the

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				manual balancing function.
cb_active_cell9	[8]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell8	[7]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell7	[6]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell6	[5]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell5	[4]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell4	[3]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell3	[2]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell2	[1]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.
cb_active_cell1	[0]	0x0	RW	After performing the read operation, "1" indicates that the battery cell is in an active balancing state. When manual balancing writes "1", it balances the specified battery cell. Writing "0" disables the manual balancing function.

3.112 CB Set LVL Register

Address Offset: 0x204

Field	Bits	Default	Type	Field Description
cb_set_lvl	[15:0]	0x0	WO	Start balancing the cells that exceed the write voltage threshold. Adjacent cells or cells that are not within the programming limits

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				will not be affected.

3.113 CB Status1 Register

Address Offset: 0x208

Field	Bits	Default	Type	Field Description
cbstatus1	[15:0]	0x0	RO	Cell Balancing Present Time

3.114 CB Status2_0 Register

Address Offset: 0x210

Field	Bits	Default	Type	Field Description
cell1_cb_time	[31:0]	0x0	RO	Cell 1 Total Balancing Time

3.115 CB Status2_1 Register

Address Offset: 0x214

Field	Bits	Default	Type	Field Description
cell2_cb_time	[31:0]	0x0	RO	Cell 2 Total Balancing Time

3.116 CB Status2_2 Register

Address Offset: 0x218

Field	Bits	Default	Type	Field Description
cell3_cb_time	[31:0]	0x0	RO	Cell 3 Total Balancing Time

3.117 CB Status2_3 Register

Address Offset: 0x21C

Field	Bits	Default	Type	Field Description
cell4_cb_time	[31:0]	0x0	RO	Cell 4 Total Balancing Time

3.118 CB Status2_4 Register

Address Offset: 0x220

Field	Bits	Default	Type	Field Description
cell5_cb_time	[31 0]	0x0	RO	Cell 5 Total Balancing Time

3.119 CB Status2_5 Register

Address Offset: 0x224

Field	Bits	Default	Type	Field Description
cell6_cb_time	[31:0]	0x0	RO	Cell 6 Total Balancing Time

3.120 CB Status2_6 Register

Address Offset: 0x228

Field	Bits	Default	Type	Field Description
cell7_cb_time	[31:0]	0x0	RO	Cell 7 Total Balancing Time

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3.121 CB Status2_7 Register

Address Offset: 0x22C

Field	Bits	Default	Type	Field Description
cell8_cb_time	[31:0]	0x0	RO	Cell 8 Total Balancing Time

3.122 CB Status2_8 Register

Address Offset: 0x230

Field	Bits	Default	Type	Field Description
cell9_cb_time	[31:0]	0x0	RO	Cell 9 Total Balancing Time

3.123 CB Status2_9 Register

Address Offset: 0x234

Field	Bits	Default	Type	Field Description
reserved	[31:0]	0x0	RO	Reserved Cell Total Balancing Time

3.124 CB Status2_10 Register

Address Offset: 0x238

Field	Bits	Default	Type	Field Description
cell10_cb_time	[31:0]	0x0	RO	Cell 10 Total Balancing Time

3.125 CB Status2_11 Register

Address Offset: 0x23C

Field	Bits	Default	Type	Field Description
cell11_cb_time	[31:0]	0x0	RO	Cell 11Total Balancing Time

3.126 CB Status2_12 Register

Address Offset: 0x240

Field	Bits	Default	Type	Field Description
cell12_cb_time	[31:0]	0x0	RO	Cell 12 Total Balancing Time

3.127 CB Status2_13 Register

Address Offset: 0x244

Field	Bits	Default	Type	Field Description
cell13_cb_time	[31:0]	0x0	RO	Cell 13 Total Balancing Time

3.128 CB Status2_14 Register

Address Offset: 0x248

Field	Bits	Default	Type	Field Description
cell14_cb_time	[31:0]	0x0	RO	Cell 14 Total Balancing Time

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3.129 CB Status2_15 Register

Address Offset: 0x24C

Field	Bits	Default	Type	Field Description
cell15_cb_time	[31:0]	0x0	RO	Cell 15 Total Balancing Time

3.130 FET Control Register

Address Offset: 0x250

Field	Bits	Default	Type	Field Description
pchg_off	[3:3]	0x0	WO	0 = PCHG FET is allowed to turn on if other conditions are met 1 = PCHG FET driver is forced off
chg_off	[2:2]	0x0	WO	0 = CHG FET is allowed to turn on if other conditions are met 1 = CHG FET driver is forced off
pdsg_off	[1:1]	0x0	WO	0 = PDSG FET is allowed to turn on if other conditions are met 1 = PDSG FET driver is forced off
dsg_off	[0:0]	0x0	WO	0 = DSG FET is allowed to turn on if other conditions are met 1 = DSG FET driver is forced off

3.131 REF12 Control Register

Address Offset: 0x254

Field	Bits	Default	Type	Field Description
reg2v	[7:5]	0x0	ROHW	Selects voltage level for REG2 0 = 3.3 V 1= 5 V 2 = 3 V 3 = 2.5 V 4 - 7 = 1.8 V
reg2_en	[4:4]	0x1	ROHW	Enables or disables REG2 0 = REG2 Disabled 1 = REG2 Enabled
reg1v	[3:1]	0x0	ROHW	Selects voltage level for REG1 0 = 3.3 V 1= 5 V 2 = 3 V 3 = 2.5 V 4 - 7 = 1.8 V
reg1_en	[0:0]	0x1	ROHW	Enables or disables REG1 0 = REG1 Disabled 1 = REG1 Enabled

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3.132 OTP WR Check FADDR Register

Address Offset: 0x258

Field	Bits	Default	Type	Field Description
otp_wr_check_faddr	[15:0]	0x0	RO	not used,always 0

3.133 OTP WR Check Register

Address Offset: 0x25A

Field	Bits	Default	Type	Field Description
ok	[7:7]	0x0	RO	This bit is set whenever programming conditions are met. None of the other bits will be set when this bit is set. 0 = OTP program not allowed 1 = OTP can program
lock	[5:5]	0x0	RO	The device is not in FULLACCESS and CONFIG_UPDATE mode, or the OTP Lock bit has been set to prevent further modification. 0 = OTP not locked 1 = OTP locked
self_chk_ok	[4:4]	0x0	RO	OTP self check ok. 0 = OTP self check error 1 = OTP self check ok
nodata	[3:3]	0x0	RO	not used,always 0
eccf	[2:1]	0x0	RO	OTP ECC Error
vpp_en	[0:0]	0x0	RO	OTP VPP enable

3.134 OTP WR FADDR Register

Address Offset: 0x25C

Field	Bits	Default	Type	Field Description
otp_wr_faddr	[15:0]	0x0	RO	OTP Write Data Fail Addr

3.135 OTP Write Register

Address Offset: 0x25E

Field	Bits	Default	Type	Field Description
otp_wr_ok	[7:7]	0x0	RO	This bit is set whenever programming success 0 = OTP programming not ok 1 = OTP programming ok
otp_wr_lock	[5:5]	0x0	RO	The device is not in FULLACCESS and CONFIG_UPDATE mode, or the OTP Lock bit has been set to prevent further modification. 0 = OTP not locked 1 = OTP locked
otp_self_chk_ok	[4:4]	0x0	RO	OTP self check ok. 0 = OTP self check error 1 = OTP self check ok

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Field	Bits	Default	Type	Field Description
otp_wr_nodata	[3:3]	0x0	RO	Could not program data into OTP (indicating data has been programmed too many times, no XOR bits left). When this bit is set, the following bytes will indicate the address of the first failing parameter. 0 = Data can be programmed into OTP 1 = Data could not be programmed into OTP
otp_eccf	[2:1]	0x0	RO	OTP ECC Error
otp_vpp_en	[0:0]	0x0	RO	OTP VPP enable

3.136 Read Cal1_0 Register

Address Offset: 0x260

Field	Bits	Default	Type	Field Description
cc2_avg_counts	[23:0]	0x0	RO	The 24-bit average value of the programmable CC2 count samples, with the sample count changed in the CC_OFFSET_SAMP Register. The highest bit is the sign bit. The unit is 28.896 nV.

3.137 Read Cal1_1 Register

Address Offset: 0x264

Field	Bits	Default	Type	Field Description
ts0_counts	[23:0]	0x0	RO	Reports the 32-bit ADC counts for the most recent measurement on the TS0 pin. When configured for thermistor measurement, the unit is 0.358uV.

3.138 Read Cal1_2 Register

Address Offset: 0x268

Field	Bits	Default	Type	Field Description
pack_pin_adc_cnt	[15:0]	0x0	RO	16-bit ADC counts from previous PACK pin voltage measurement.

3.139 Read Cal1_3 Register

Address Offset: 0x26A

Field	Bits	Default	Type	Field Description
top_stack_adc_cnt	[15:0]	0x0	RO	16-bit ADC counts from previous Top of Stack voltage measurement.

3.140 Read Cal1_4 Register

Address Offset: 0x26C

Field	Bits	Default	Type	Field Description
ld_pin_adc_cnt	[15:0]	0x0	RO	16-bit ADC counts from previous Top of LD pin measurement

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3.141 Security Access Register

Address Offset: 0x280

Field	Bits	Default	Type	Field Description
access_key_step2	[31:16]	0xff	RWAC	Access Key Step 2
access_key_step1	[15: 0]	0xff	RWAC	Access Key Step 1

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4 Revision History

REVISION NUMBER	DESCRIPTION	DATE
A	Initial release	April 29, 2025
B	Updated units, balancing, and alarm status,	Dec 15, 2025
C	Updated current and IADC unit	Mar 30, 2026
D	Correct address errors, Update the description of the temperature register	Apr 15, 2026

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